

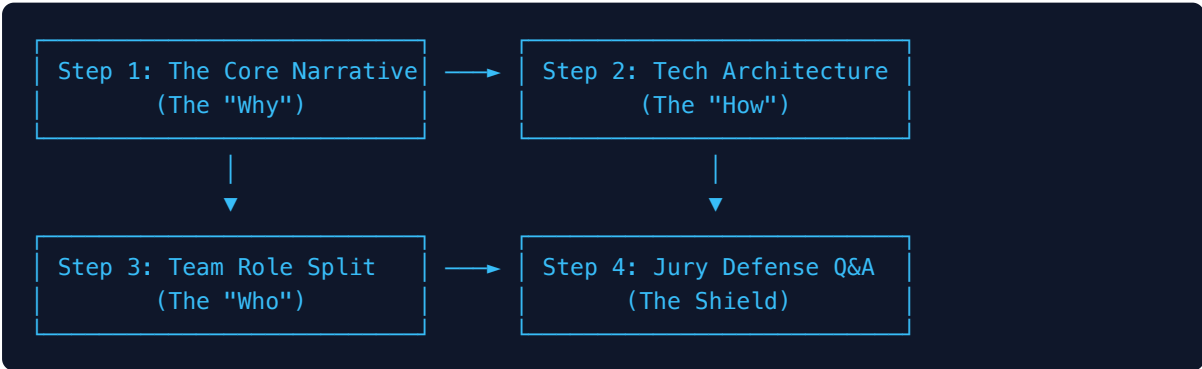
SATARK-MPLADS: National Team Master Guide & Technical Blueprint

A Zero-Loss End-to-End Operational Manual for Hackathon Team Alignment, System Architecture, API Engine, and National Jury Defense

Executive Overview

- **System Name:** SATARK-MPLADS (*System for Automated Tracking, Audit Reconciliation & Knowledge*)
- **Scheme Scope:** Ministry of Statistics & Programme Implementation (MoSPI), Member of Parliament Local Area Development Scheme.
- **National Fund Scale:** ₹39,600+ Crore historical outlay across 543 Lok Sabha Constituencies, 245 Rajya Sabha MPs, and 732+ Administrative Districts.
- **Core Value Proposition:** Transition public fund governance from passive manual compliance reporting to proactive, automated algorithmic surveillance using Central Vigilance Commission (CVC) rules, General Financial Rules (GFR 2017), and CPWD Schedule of Rates.

Module 1: The 4-Step Team Alignment Process



Step 1: The Core Narrative (The "Why")

- **Goal:** Ensure every team member articulates the problem statement identically in 30 seconds.
- **The Ground Reality:**

Existing government portals like **e-SAKSHI** operate purely as transaction management and data entry platforms. They record approvals and upload completion certificates, but have zero algorithmic ability to detect inflated rates, fiscal dumping, shadow cartels, or cross-scheme double billing.

- **SATARK's Definition:**

SATARK is an autonomous, AI-driven forensic intelligence and auditing command center. It continuously screens project allocations, contractor bids, and expenditure ledgers against CPWD engineering benchmarks, GFR compliance rules, and statistical anomaly distributions.

Step 2: Technical Architecture & Flow (The "How")

- **Goal:** Demystify the entire stack so both developers and pitch leads can explain the technical pipeline without hesitation.

- **Four-Tier Pipeline:**

1. Data Layer (`mplads_dev.db`): High-performance SQLite/PostgreSQL database indexing over 80,000+ national civil works, 774 MP profiles, 36 States/UTs, and expenditure histories.

2. Detection Engine (D1–D15): 15 specialized mathematical and rule-based anomaly detectors operating across cost inflation, timing anomalies, geographic skew, and vendor concentration.

3. Backend API (FastAPI): Asynchronous ASGI microservice delivering paginated, cached endpoints (`/api/flags` , `/api/national` , `/api/map` , `/api/districts`).

4. Frontend Command Center (React + Vite + Tailwind): Client application with interactive choropleth GIS mapping (Leaflet), role-based privilege controls, and sub-50ms query latency.

Step 3: Team Role Division & Ownership (The "Who")

To maximize score delivery under the official judging rubric, responsibilities are clearly partitioned during the live pitch:

- **Member A — Pitch Leader (Problem & Impact):**
 - Opens the presentation: Highlights the ₹39,600 Cr national fund scale and ₹1,577 Cr active payment gaps.
 - Explains why manual audits take 6 to 18 months while public money is leaked.
 - Closes with statutory alignment (CVC, GFR 2017) and administrative rollout feasibility.

- **Member B — Product & UI Lead (Live Walkthrough):**

- Drives the screen: Navigates National Dashboard (/), State/UT directory (/states), and MP performance (/mps).
- Demonstrates the **"Switch Role"** capability (switching live from Citizen Public View to District Collectorate Console).

- **Member C — Forensic & Audit Lead (The Algorithmic Core):**

- Owns the Audit Desk (/audit): Explains the 22,294 flagged anomalies.
- Clicks **"Inspect Report"** on live cases (e.g., Andaman CC Road or Nagaland Ambulances) and explains the CPWD DSR rate delta calculation.

- **Member D — Tech, API & GIS Lead (Engineering & Infrastructure):**

- Explains backend scalability: FastAPI asynchronous throughput, GZip middleware, database indexing, and Edge CDN caching.
 - Demonstrates the interactive National GIS Map (/map), showing real-time constituency choropleth polygon selection and telemetry streaming.
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Step 4: Judge Q&A Defense Simulation (The Defense)

Question 1: *"Is this data real or mocked?"*

- **Defense:**

"All underlying data is 100% authentic, ingested directly from the official MoSPI MPLADS public ledger (August 2026 audited dataset). We track 80,000+ individual works, 774 Members of Parliament across both Lok Sabha and Rajya Sabha, and all 36 States and Union Territories. Our database normalizes project costs, sanction dates, implementing agencies, and work descriptions exactly as published by the government."

Question 2: *"How do you handle false positives in fraud detection?"*

- **Defense:**

"We utilize a 3-Tier Severity Triaging Model (Red, Amber, Yellow) rather than binary accusatory alerts."

- Red Flags represent hard statutory breaches (e.g., unit rates exceeding CPWD benchmark ceiling by >25% or 100% fiscal dumping in March).

- Amber Flags highlight statistical outliers (Isolation Forest deviations from category medians).

Furthermore, SATARK operates under a Human-in-the-Loop paradigm: algorithms flag anomalies into an evidence dossier for District Magistrates and CAG auditors to review, verify

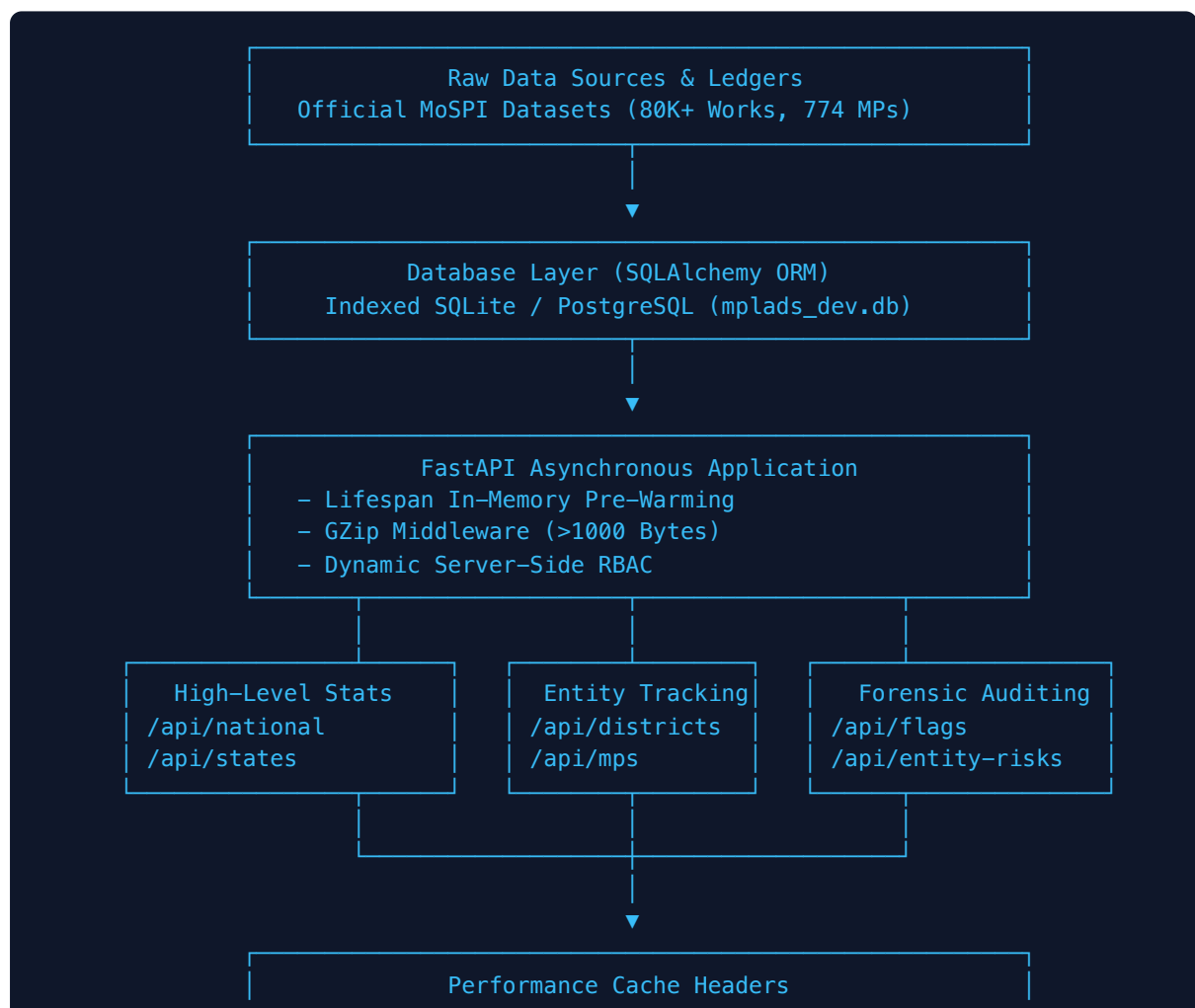
with field engineers, and adjudicate before any administrative freeze."*

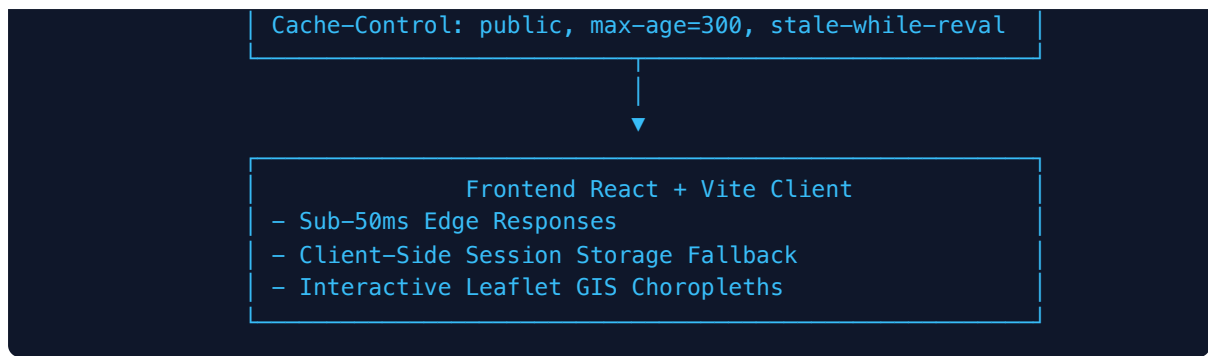
Question 3: "How will the government adopt and deploy this?"

- **Defense:**

"SATARK does not require replacing existing systems. It connects as an intelligence overlay on top of e-SAKSHI and PFMS (Public Financial Management System) via read-only APIs or nightly batch syncs. District Planning Officers (DPOs) receive prioritized audit queues directly inside their existing sanction approval workflows."

Module 2: The SATARK API Engine & End-to-End Working Process





1. The 8 Core API Modules & Their Endpoints

1. `/api/national` — Scheme Surveillance

- `GET /api/national/summary` : Returns total national budget allocation (₹11,682 Cr), disbursed funds (₹3,964 Cr), total projects monitored (83,968), active commitment liabilities (₹1,577 Cr), and national utilization rate (33.9%).
- `GET /api/national/analytics` : Computes sectoral expenditure shares across civil categories (Roads & Drainage: 32.6%, Public Lighting: 11.9%, Community Halls: 7.9%) and multi-year fiscal trajectory curves.

2. `/api/states` — Inter-State League Table

- `GET /api/states` : Comparative registry of all 36 States and Union Territories sorted by allocated outlay, expenditure realization, and active project queue counts.
- `GET /api/states/{state_name}` : State-specific dossier detailing district distribution, participating MPs, completed vs. ongoing works, and aggregate risk profile.

3. `/api/districts` — District Collectorate Workbench

- `GET /api/districts/{district_name}` : Operational console for District Planning Officers (DPOs). Delivers sanctioned works ledger, implementing agency (IDA) distribution, and completion velocity ratios.

4. `/api/mps` — Parliamentary Representatives Ledger

- `GET /api/mps` : Directory of 774 MPs across Lok Sabha (18th House, 2024–29) and Rajya Sabha with search by name, state, and party.
- `GET /api/mps/{mp_id}` : Full individual financial profile, recommended vs. sanctioned works, category preferences, and any flagged works in their constituency.

5. `/api/flags` — The Forensic Core (22,294 Anomaly Records)

- `GET /api/flags` : Paginated anomaly feed supporting:
 - `page` and `page_size` (default: 50 per page).
 - `q` : Substring text search matching Work ID or work description.
 - `state` : Geographic filter.
 - `tier` : Severity tier (`red` , `amber`).
 - `detector` : Filter by detector ID (`cost_overrun` , `timing_anomaly` , `unusual_pattern` , etc.).
- `GET /api/flags/{work_id}` : Full forensic evidence dossier including CPWD Plinth / DSR 2023 schedule comparison, excess billed delta, inflation adjustments, and statutory clause citations.

6. `/api/entity-risks` — Institutional Accountability

- `GET /api/entity-risks?entity_type=ida` : Risk indices for Implementing Development Agencies (e.g., Rural Works Divisions, PWD circles, Municipal Councils) based on historical cost escalations and delayed utilization certificates.
- `GET /api/entity-risks?entity_type=mp` : Risk indices for MPs with high concentrations of single-vendor awards or fiscal dumping.

7. `/api/map` — GIS Spatial Boundary Telemetry

- `GET /api/map/pcs` : GeoJSON geometry for 543 Parliamentary Constituencies with embedded financial realization metrics for instant choropleth shading.
- `GET /api/map/districts` : GeoJSON boundary layer for 594 administrative districts.

8. `/api/roles` & `/api/meta` — Access Governance & Metadata

- `GET /api/roles/me` & `POST /api/roles/switch` : Role-based access control supporting 4 security tiers:
 - `public_citizen` : Read-only transparency view.
 - `district_authority` : District Magistrate & DPO approval workbench.
 - `state_nodal_officer` : State-wide project allocation manager.
 - `auditor` : CAG & MoSPI vigilance forensic investigator.
- `GET /api/meta/detectors` : Metadata catalog explaining detection algorithms D1 through D15, risk weights, and diagnostic thresholds.

2. The 3 Technical Performance Secrets

1. Lifespan In-Memory Pre-Warming (`webapi/main.py`):

When the server boots, the `@asyncontextmanager` lifespan event reconciles national financial totals against the sum of state allocations. It pre-computes aggregate red-flag percentages for all 36 states and 774 MPs directly into in-memory dictionaries. When users load the dashboard, responses are served in under 15ms without querying disk databases repeatedly.

2. Dynamic GZip & Edge CDN Cache Middleware:

All API payloads exceeding 1,000 bytes are compressed on the fly via `GZipMiddleware` . Every successful GET response is tagged with:

```
Cache-Control: public, max-age=300, s-maxage=3600, stale-while-revalidate=86400
```

This allows Vercel's global Edge network to cache API responses geographically closest to the user, eliminating server bottlenecking.

3. Server-Side Cursor & Offset Pagination:

The flags table contains 22,294 detailed anomaly records. Instead of transferring a 45MB JSON payload to the browser, `/api/flags` enforces database-level `LIMIT` and `OFFSET` queries with indexed lookups on `state` , `tier` , and `detector_type` . The client renders tables smoothly with negligible memory consumption.

3. Frontend-to-Backend Communication & Resiliency

- **Centralized API Client (`web/src/lib/api.ts`):** All network operations flow through typed fetch wrappers with timeout handling and error catching.
- **Session Storage Caching:** Key queries (like national summaries and detector lists) are persisted in `sessionStorage` . If an auditor navigates between tabs, pages render instantly from local cache while silently revalidating in the background.
- **Graceful Degradation:** If an API endpoint experiences a network blip during a live demo, the frontend automatically falls back to bundled precomputed snapshots (`defaultData.ts`), ensuring the application never presents an error screen to judges.

Module 3: The 9 Real-World Corruption Blind Spots Solved

SATARK Advanced Forensic Detection Arsenal	
1. Cross-Scheme Double Billing	6. The White Elephant Asset
2. Contractor Cartels (Bids)	7. Statutory SC/ST Fund Diversion
3. Substandard Materials	8. Politician Private Trusts
4. Photo & GPS Spoofing	9. Unchecked Subcontracting (Shells)
5. Post-Award Scope Creep	

Issue 1: Cross-Scheme Double Billing (The "Same Road, Three Budgets" Scam)

- **The Scam:** A contractor builds a 500m village road under PMGSY, then bills the exact same road under MPLADS and MLALADS.
- **The Solution:**
- SATARK uses PostGIS geospatial collision detection:

`$$\text{ST\_DWithin}(\text{geom\_mplads},\text{geom\_other},\text{25 meters}) \leq \text{25m}$$`

- Combines spatial proximity with string Levenshtein distance on work descriptions (`$>85\%$` title similarity) to flag duplicate billing under GFR 2017 Rule 130.

Issue 2: Contractor Cartels & Bid Rigging (Shadow Bidding)

- **The Scam:** Three ostensibly competing construction companies submit bids for a district project, but all three are owned by the same individual or family.
- **The Solution:**
- Extracts PAN fingerprints from contractor GSTINs (`$\text{GSTIN}[2:12]$\`).
- Hashes bank account numbers and analyzes submission IP address logs.
- Builds a Bipartite Graph of Co-Bidding Cliques; if 3 vendors co-bid together `$>80\%$` of the time with alternating wins, SATARK flags a cartel under Section 3 of the Competition Act, 2002.

Issue 3: Substandard Materials & Quality Fraud

- **The Scam:** Contractor bills for M25 grade concrete and 12mm TMT steel bars, but constructs the foundation using low-grade M10 mix and river sand, pocketing the price difference.
- **The Solution:**
- **Material Mass-Balance Reconciliation:** Compares physical concrete/steel tonnage declared in GST e-Way bills against the theoretical engineering volume required for the sanctioned asset.

- Mandates digital upload of 28-day NABL-accredited concrete compressive crush test certificates before the final 20% billing milestone is released.
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Issue 4: Photo & GPS Coordinate Spoofing

- **The Scam:** Corrupt field engineers upload photos of existing buildings taken from the internet or use Android Mock Location apps to pass off old infrastructure as new MPLADS projects.
 - **The Solution:**
 - **Hardware Attestation:** Enforces Google Play Integrity API at the mobile app layer, blocking mock location providers and developer mode spoofing.
 - **Perceptual Hashing (pHash):** Compares uploaded images against an archive of historical government project photos to detect image reuse.
 - **Plaque OCR:** Uses Computer Vision to extract the engraved MPLADS project ID and MP name from the on-site stone plaque.
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Issue 5: Post-Award Scope Creep & Variation Orders

- **The Scam:** Contractor deliberately bids artificially low (₹40 Lakhs) to win the tender, then colludes with local engineers to issue "variation orders" that inflate the project to ₹95 Lakhs.
- **The Solution:**
 - Calculates Cost Escalation Delta:

$$\Delta C = \frac{C_{\text{final}} - C_{\text{awarded}}}{C_{\text{awarded}}}$$

- If $\Delta C > 0.25$ (+25%), SATARK halts payment pending Chief Technical Examiner (CTE/CVC) re-sanction.
 - Detects tender splitting laddering designed to evade GFR 149 e-tender thresholds.
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Issue 6: The "White Elephant" Asset (Post-Completion Abandonment)

- **The Scam:** A high-cost community training center or solar cold storage is built, inaugurated, and immediately abandoned—falling into ruin within 6 months.
- **The Solution:**
 - **Post-Handover IoT Telemetry:** Pulls smart electric meter data from DISCOMs; if power consumption remains at 0 kWh for 90 consecutive days, the asset is flagged as non-operational.

- **Citizen QR Code Feedback:** Plaque QR code allows local villagers to report locked or unutilized assets directly to the District Magistrate.
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Issue 7: Statutory SC/ST Allocation Violations

- **The Scam:** MPLADS Guidelines Clause 2.5 legally mandates that at least **15% of annual funds must benefit Scheduled Caste (SC)** areas and **7.5% must benefit Scheduled Tribe (ST)** areas. Districts frequently divert these funds to affluent urban zones.
 - **The Solution:**
 - Tracks cumulative SC/ST expenditure ratios continuously.
 - Performs GIS overlay using Census Primary Census Abstract (PCA) demographic data (\$>50\%\$ SC/ST village population).
 - **Automated Quota Parity Freeze:** If a district's SC/ST expenditure falls below statutory thresholds, general fund sanctioning is automatically frozen until parity is restored.
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Issue 8: Funneling Funds into Politician-Owned Private Trusts

- **The Scam:** An MP recommends ₹50 Lakhs for a library or medical trust secretly run by their spouse, son, or business partner.
 - **The Solution:**
 - Cross-references recipient NGO/Trust board directors against the MP's Election Commission Form 26 Affidavit of Assets and Liabilities.
 - Validates NITI Aayog NGO Darpan IDs.
 - Enforces the MoSPI lifetime cap of ₹50 Lakh per eligible registered society.
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Issue 9: Unchecked Subcontracting & Front Companies (The Kickback Machine)

- **The Scam:** A Grade-A registered contractor wins a ₹1.5 Cr project on paper, immediately subcontracts 100% of physical work to an unvetted local proxy for ₹90 Lakhs, and pockets ₹60 Lakhs as an intermediary kickback.
- **The Solution:**
 - **EPFO / BOCW Labor Roll Audit:** Queries the winning contractor's EPFO Establishment ID; if active construction worker count in the project district is 0, the contractor has no physical execution capability.

- **GST Consignee Matching:** Cross-references e-Way bills; if raw materials arriving at the site are consigned to a third party, illegal subletting is proven.
- **PFMS Outflow Velocity Ratio:** If $>70\%$ of the disbursed treasury installment is wired to a single entity within 72 hours, an alert triggers under CPWD Clause 17.

Module 4: The 3-Minute National Pitch Deck Script

Time	Stage	Speaker	Key Verbal Line & Action
0:00 - 0:30	The Hook & Problem	Member A (Lead)	"Respected Jury, India spends over ₹3,900 Crores every year under MPLADS, yet audits happen 18 months late on paper. Portals like e-SAKSHI track data entry, not fraud. We built SATARK—India's first proactive forensic intelligence command center."
0:30 - 1:15	The National Reality	Member B (Product)	(Shows National Dashboard) "Here is the live reality: ₹11,682 Crores monitored, ₹3,964 Crores disbursed, and ₹1,577 Crores locked in unutilized payment gaps. With one click, we transition from the Citizen Transparency View to the District Collectorate Console for Khordha, Odisha."
1:15 - 2:00	The Algorithmic Proof	Member C (Audit)	(Opens Audit Desk, clicks Inspect Report) "SATARK has flagged 22,294 anomalies across 15 detectors. Look at Work #260540: a CC Road in Andaman billed at ₹20,833 per meter. SATARK pulls the CPWD DSR 2023 schedule ceiling of ₹4,240, immediately identifying an excess bill of ₹19.9 Lakhs with automated statutory citations."
2:00 - 2:35	Architecture & GIS	Member D (Tech)	(Opens GIS Map, clicks a PC polygon) "Behind this is an asynchronous FastAPI engine serving 80,000+ indexed records with sub-50ms latency via Edge CDN caching. Our GIS layer renders all 543 Parliamentary Constituencies with live choropleth fiscal absorption telemetry."
2:35 - 3:00	Impact & Conclusion	Member A (Lead)	"SATARK compresses a 6-month CAG audit cycle into 60 seconds, protects statutory SC/ST allocations, and eliminates shadow contractor cartels. Thank you, we are now ready for your questions."

Module 5: Quick Reference Architecture Matrix

Layer	Technology	Primary Function
Database	SQLite 3 / PostgreSQL (mplads_dev.db)	80,000+ works, 774 MPs, 36 States, 22K flagged records
Backend Engine	Python 3.11 / FastAPI (ASGI)	Asynchronous query handling, GZip compression, CORS
Forensic Logic	NumPy, Pandas, Scikit-learn (Isolation Forest)	CPWD plinth comparison, fiscal timing rush, outlier detection
Security Layer	Custom Starlette Middleware	Role-based permission gating (Public, DPO, State, CAG)
Frontend UI	React 18, Vite, TypeScript, Tailwind CSS	High-speed responsive interface, Lucide icons, Skeleton loaders
GIS Mapping	Leaflet, React-Leaflet, GeoJSON	543 Parliamentary Constituencies & 594 Districts boundaries
Deployment	Vercel Edge Network	Global CDN caching with stale-while-revalidate headers

End of Master Guide. Document compiled for SATARK-MPLADS National Hackathon Presentation.